

Google's Internet balloons

Geo-stationary balloon satellites (GBSes) float in the atmosphere, 18km to 21km above sea level at a fixed point. Air density at that height is one-tenth of sea level. Average wind speed there is also low. A propulsion system moves and maintains the balloons in position.

GBSes are powered by solar panels and receive laser power from the cell tower these hover over. These could be used to provide broadband Internet access over a large area.

To provide Internet connectivity at remote places, Google came out with an innovative solution—known as Project Loon—using such balloons. It consists of a network of balloons floating in the stratosphere. Balloons have transceivers that transmit and receive signals. The system utilises the wind obtained from National Oceanic and Atmospheric Administration (NOAA), and joins the global network by establishing a connection using Internet Service Protocols (ISP) or Long-Term Evolution (LTE) technologies. Loon balloons can travel in different directions.

Each balloon contains the following:

- An envelope made from polyethylene plastic: It is 15m wide and 12m tall. It is a super-pressure, specially-constructed balloon, which lasts longer than a weather balloon, and can withstand higher pressure.
- Solar panels that provide power to the balloons: These sit between the envelope and equipment. Under full Sun, panels get fully-charged and produce around 100W of power. This power is enough to keep the unit in service mode, while charging the battery for use at night. Project Loon gets power from wind, too.

- A box containing electronic components: The box looks like a basket. It contains electronic circuit to controls the system, radio antennae for communication with other balloons and the Internet antennae on ground. It also has a battery that acts as a power backup unit that stores solar power so that balloons can operate at night, too.

Advantages of Loon.

- Is cost effective
- Improves communication during disaster
- Uses renewable energy sources and is available at all times and places
- Is fast, efficient and more reliable than wired connections

Disadvantages.

- Hardware failure
- Is restricted by spectrum regulations

Mobile towers, TVs and the IoT for rural India

Many telecom giants place mobile towers only in cities and do not focus on rural areas. This is due to huge costs incurred in the setting up of these towers. Today, huge subsidies are being provided by USOF to set up passive infrastructures such as mobile towers and power supplies (such as diesel generators). Subsidies are also being provided to install active telecommunication equipment like dish antennae, flat antennae and backhaul carriage facilities.

Telecom operators continue to face problems in installing mobile towers in rural sectors. This is because individual states impose their own rules that do not match the rules imposed by Department of Telecom (DoT). Tower and Infrastructure Providers Association is fighting to overcome these differences.

High competition exists in the field of direct-to-home (DTH) set-top box market in India. DTH operators provide very-high-speed broadband in Ka band, with 50mbps downlink and 20mbps uplink speeds.

Today, 70 per cent of DTH subscribers reside in rural areas. Tata Sky is the dominant DTH brand with 13.2 million DTH subscribers from rural areas alone. The rural electrification initiative by the government of India is the main reason behind the growth of Internet technology in rural India.

Virtual network operators is another recent trend. These provide services in rural areas using a network of existing TSPs having unutilised capacity or using last-mile connectivity.

National Telecom Policy 2018 aims to provide new technologies including 5G and the Internet of Things (IoT) to rural areas. Foreign direct investments also play a crucial role in the development of the telecom sector in rural areas.

Rural farming has evolved through initiatives like subsistence farming, large for-profit farms, animal-powered farming and machine farming. We are now entering the era of data-powered farming via large-scale open data sets on weather and yields, the IoT, drones and smart tractors. With help from India Electronics and Semiconductor Association (IESA), several startups are developing hardware and software to assist rural farming.

Today, small-scale farmers use drip irrigation particularly in areas where there is shortage of water and water is a precious resource. These can now be controlled remotely via a combination of installed IoT sensors and activators along with mobile applications. **EFY**

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